



GEOptimize Inc
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Proposal for:
Greg Rogers
Chief Executive Officer
UM Properties GP Inc.
100 Innovation Dr.
Winnipeg, MB
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Proposal Date:
2022-10

Proposal Number:
QU-056

Reference
CA MB UM Properties – Phase 1A

Understanding of project

UM Properties GP is the land developer of the former Southwood golf course in Winnipeg. The land area available totals approximately 112 acres and has been approved under a secondary plan for approximately 10,000 multifamily residential units. A district geothermal system has been proposed for Phase 1A of this development which is comprised of 2,000 residential units in Southwood Circle community. UM Properties is requesting pricing to design, build and operate for the Phase 1A district geothermal system.

Phase 1A includes approximately 3,000,000 ft² of floor area with 80% multifamily residential units and 20% retail units. The buildings will be designed to meet LEED Gold rating. For the purpose of this feasibility study, it will be assumed that Phase 1A consists of eight 500,000 ft² buildings each with 20 floors at 25,000 ft² per floor. The insulation will be R-27 for the walls and R-40 for the roof with double pane glass at 30% window to wall ratio. The building energy models will include energy recovery ventilator units with a 90% efficiency.

Proposed design methodology

District Geothermal System Modelling

Energy loads of individual buildings are calculated based on the proposed architectural and mechanical system design, anticipated occupancy, and the historical hourly weather data of the location.

GEOptimize will employ hourly energy models of representative buildings similar to those proposed by UM Properties GP. The energy model geometry will be based on a prototypical building type from the GEOptimize energy model library created using Trane Trace 700 and simulated with Winnipeg, Manitoba weather data. The prototypical energy models will be manipulated to match the information given for this project as described in the Section 'Understanding of project'. GEOptimize may recommend the integration of energy efficiency measures that can be added to the buildings to balance the annual heating and cooling energy loads. Balancing annual energy loads will reduce the size and

cost of ground heat exchanger for the system and reduce the potential of long-term temperature degradation of the ground heat exchanger or groundwater system that is required for the proposed ground source heat pump system.

The type and configuration of ground heat exchanger (GHX) that proposed for the district geothermal system will be determined by evaluating the least capital-intensive option. A KGS feasibility study determined that the target flow rates for the UM Lands Phase 1A development are 120 to 180 gpm from each of 20 to 30 wells which totals a flowrate of approximately 3,600 gpm. The study identified the possibility of extracting energy from the Red River and storing the heat in the aquifer during the summer to mitigate the potential of long-term reduction of the aquifer and improve the efficiency of heat pumps connected to the system during the heating season. GEOptimize will investigate and include these options in the conceptual design for the purpose of the feasibility study.

GEOptimize will develop a preliminary design of a 5th Generation (5G) district geothermal system using TRNSYS V18.03. TRNSYS is a simulation program used to model large, interconnected energy systems. Energy loads for the eight individual buildings will be imported into the TRNSYS model to simulate the building loads and their impact on the district energy system. The software is then used to analyze the relationship between the GHX's, buildings, and other potential energy resources. Other energy sources and / or heat sinks that can be modelled with TRNSYS include wastewater heat transfer, snow melt system heat dissipation, open water well heat exchangers, river heat exchangers, and ice rink heat injection such as the nearby Max Bell Centre ice arena. GEOptimize will also analyze the addition of a snow melt system for a 1 km street on the University campus. UM Properties GP will supply GEOptimize with the estimated kBtu's and GEOptimize will include this in the TRNSYS model. The feasibility of the snow melt system will be analyzed as well as the impact of the size and cost of the 5G district GSHP system.

Several alternatives of the TRNSYS model will be developed to optimize the system performance including:

- Size of GHX modules and / or water wells
- Type of GHX modules
- Location of GHX modules connected to the district energy transfer pipe
- System control strategy
- Additional waste and renewable heat sources/sinks that may be identified during the investigation

Capital will be estimated in collaboration with Postma cost consultanting. The capital cost estimate will be performed at a Class D estimation level. Operating costs will be estimated based on the applicable utility rate structure and the hourly energy models developed for the investigation. The modularity of the 5th generation DES allows for a reduction of initial capital cost investment as the project can be broken into steps as each new building is added onto the district loop. The developer can build the ETP infrastructure with the first building and connect more buildings onto the ETP as fast or as slow as they'd like. This creates flexibility in capital allocation for the developer while energy savings are compounded as more buildings are connected to the ETP.

Scope of work

- Overview of suitable types of ground heat exchangers.
- Identification of possible heat sources/sinks to reduce capital and operating cost of the 5G district system.
- Develop eight energy models based on the supplied information and manipulating a prototypical model from the GEOptimize energy model library.
- Preliminary conceptual design of the 5G district energy system to meet the building heating and cooling demand based on the developed energy models including:
 - Preliminary conceptual design of a district energy system demonstrating the modular nature of 5G district system can grow and evolve with the development and how it can potentially be expanded to include existing buildings on the U of M campus
 - Capacity of heat pump equipment required for each building
 - Size and length of energy transfer pipe infrastructure
 - Energy transfer pipe pump size and control sequence
 - Conceptual system sequence of operation
 - Size and configuration of ground heat exchanger modules
 - Potential of other waste energy and renewable energy resources
- Develop a Class D level capital and operating costs estimate in collaboration with a cost consultant based on the preliminary design.

Deliverables

GEOptimize will prepare a PowerPoint presentation to be delivered to UM Properties GP at a meeting via screen sharing software detailing the system design and costs. GEOptimize will also provide a brief report summarizing the work completed.

Pricing

Description	Rate	Qty	Line Total
Preliminary District GSHP System Design	19,750.00	1	19,750.00
3 rd Party Cost Consultant (Postma Consulting)	4,000.00	1	4,000.00
Subtotal			23,750.00

Proposal Total (CAD)

\$23,750.00

Terms

- Additional work out of scope will be charged at \$250/hr.
- Invoices will be submitted monthly for work completed to date. Payment is expected on receipt of invoice.

Authorized signature – UM Properties GP Inc.